



Department of Information Technology

Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch : IV Semester-B. Tech Information Technology

Course Code & Title : IT3401 & Web Essentials (Integrated Course)

Name of the Faculty member (s) : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic** : Unit 3 / Form Validation
- **Course Outcome** : CO3
- **Topic Learning Outcome:** PrO3
- **Activity Chosen** : Think Pair Share
- **Justification:**
 - Form validation plays a vital role in interactive web application development by ensuring that user input is checked on the client side before being sent to the server.
 - To enhance students' understanding of this concept, the Think-Pair-Share activity has been selected. This method encourages students to explore and discuss the topic collaboratively, enabling them to effectively write regular expressions for validating common form fields such as email addresses, passwords, and mobile numbers.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - First, the instructor introduced students to the syntax of regular expressions, including modifiers and commonly used metacharacters.
 - Following this, the instructor explained how to construct regular expressions to validate various form fields such as email, password, and mobile number, using a real-time example of a student registration form.
 - After the explanation, a Think-Pair-Share activity was conducted for a duration of 20 minutes to reinforce the concept and encourage collaborative learning among students.
 - The students were instructed to think individually and write regular expressions for validating password patterns, email IDs and mobile numbers based on their own form designs within 5 minutes, as shown in Figure 1.
 - Following the individual activity, students were asked to collaborate with a partner or form small groups with nearby classmates to discuss the regular expressions they had written for form validation. They spent 5 minutes reviewing and justifying their solutions with one another, as depicted in Figure 2.

- At the end, the instructor, asked a few pairs of students to come to the front of the class and present their validation code for the form design as depicted in Figure 3.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO3
CO3	2	1	2	2	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO3
Justification for correlation	Students will be able to understand the fundamental concepts of regular expressions and form validation techniques to solve real-world web input challenges.	Students will be able to analyze user input problems in web forms and identify the need for specific validation rules	Students will be able to design efficient solutions for validating web form controls using regular expressions	As students engage in the activity as a team, it fosters both independent thinking and collaborative skills, helping them work effectively on form validation using regular expressions.	Students communication skills were enhanced as they explained their form validation solutions to their peers.	Students will develop an understanding of how input validation enhances security, ensures data integrity, and improves user experience in software projects

- **Images / Screenshot of the practice:**



Fig 1: Students thinking about the given problem



Fig 2: Pair with neighbour and discuss answer

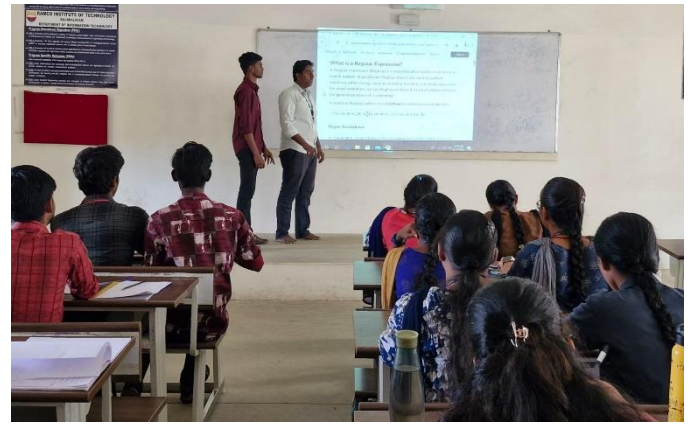


Fig 3: Presentation by R. Sibiah and M. Jeeva Sri Team R. Dharani balan & S. Gopi Team

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students expressed that the Think-Pair-Share activity significantly improved their understanding of real-time form validation using JavaScript and regular expressions.
- After the activity, Students know how to construct regular expressions to handle different input patterns and requirements like enforcing strong password rules.

- ❖ ***Benefit of the practice:***

- The Think-Pair-Share activity encourages students to independently explore strategies for implementing form validation and solving problems using JavaScript and regular expressions.
- This activity promotes active participation by engaging every student with the topic, instead of relying on passive listening during lectures.

- ❖ ***Challenges faced in implementation:***

- Students found it hard to relate validation logic to real-world use cases without practical examples.
- Some students were confused about the differences between client-side and server-side validation

Some students struggled with writing validation logic, as observed. To support them, the instructor walked through the syntax and logic of regular expressions for validating form fields, using practical examples.

References:

- https://www.ritrjpm.ac.in/images/B.Tech-IT/2023-2024/IP/2_CS3352_VA_TPS.pdf

Signature of Faculty Member

HOD